Banking Fraud Detection

Data Warehouse Schema Design

1. Design Approach & Architecture

This data warehouse follows the Kimball dimensional modeling methodology, organized as a star schema. The model is purpose-built to support three analytical domains: transaction analytics, fraud detection & ML model performance monitoring, and banking operations (loans, accounts, branches).

The schema is divided into two layers:

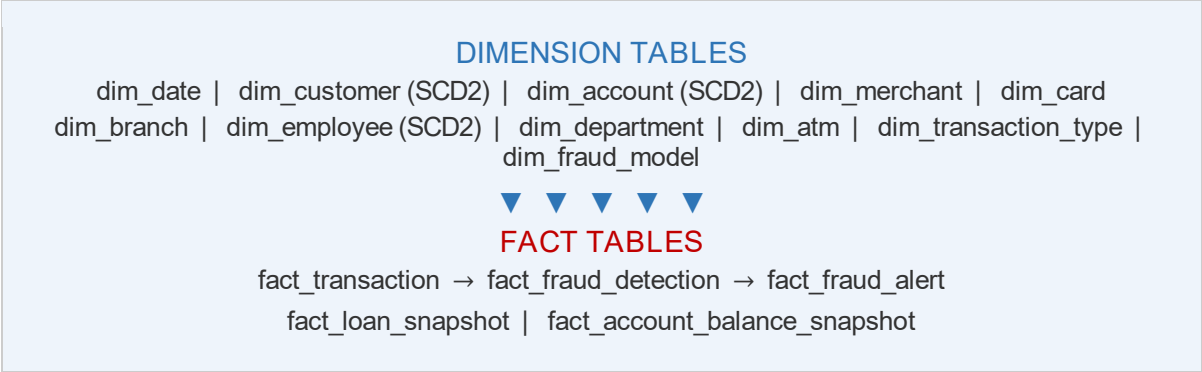
Dimension Tables (DIM_*): Descriptive context entities — Customer, Account, Merchant, Card, Branch, Employee, ATM, Date, TransactionType, and FraudModel. Slowly Changing Dimension Type 2 (SCD2) is applied to Customer, Account, and Employee to preserve historical states for point-in-time accuracy.

Fact Tables (FACT_*): Measurable events — Transactions, Fraud Detection Results, Fraud Alerts, Loan Snapshots, and Account Balance Snapshots. All facts reference dim_date as the primary time axis.

Key design decisions: (1) FraudAlert and FraudDetectionResult are modeled as fact tables in the DWH — they are output artifacts of the ML pipeline, not source OLTP records. (2) Card numbers are stored as SHA-256 hashes to maintain PCI DSS compliance. (3) All monetary amounts include a normalized USD column for cross-currency analysis. (4) The dim_fraud_model table enables multi-model comparison and champion/challenger tracking.

2. Star Schema Overview

The diagram below summarizes the relationships. Each fact table sits at the center with dimension tables radiating outward. fact_fraud_detection bridges fact_transaction and the ML output layer.



3. Dimension Tables

3.1 dim_date

Universal time dimension shared by all fact tables. Populated once and never changed (Type 1). The surrogate key uses YYYYMMDD integer format for fast range scans (e.g. WHERE date_key BETWEEN 20240101 AND 20241231).

Key	Column Name	Data Type	Nullable	Description
PK	date_key	INT	NO	Surrogate key (YYYYMMDD format)
	full_date	DATE	NO	Full calendar date
	day_of_week	VARCHAR(10)	NO	Monday, Tuesday, etc.
	day_of_week_num	TINYINT	NO	1=Monday … 7=Sunday
	day_of_month	TINYINT	NO	1–31
	day_of_year	SMALLINT	NO	1–366
	week_of_year	TINYINT	NO	ISO week number 1–53
	month_num	TINYINT	NO	1–12
	month_name	VARCHAR(10)	NO	January, February, etc.
	quarter_num	TINYINT	NO	1–4
	year_num	SMALLINT	NO	Calendar year e.g. 2024
	is_weekend	BOOLEAN	NO	TRUE if Saturday or Sunday
	is_holiday	BOOLEAN	NO	TRUE if public holiday
	fiscal_quarter	VARCHAR(6)	YES	Fiscal quarter label e.g. FY25Q1

3.2 dim_customer (SCD Type 2)

SCD2 is applied to preserve historical customer attributes. When a customer changes address, segment, or occupation, a new row is inserted with a new effective_date and the old row's expiry_date is set. is_current=TRUE always identifies the latest version.

Key	Column Name	Data Type	Nullable	Description
PK	customer_key	INT	NO	Surrogate key (DWH generated)
FK	customer_id	VARCHAR(50)	NO	Business key from source Customer table
	first_name	VARCHAR(100)	NO	
	last_name	VARCHAR(100)	NO	
	email	VARCHAR(200)	YES	
	date_of_birth	DATE	YES	
	gender	VARCHAR(20)	YES	
	marital_status	VARCHAR(20)	YES	
	occupation	VARCHAR(100)	YES	
	city	VARCHAR(100)	YES	

Key	Column Name	Data Type	Nullable	Description
	state	VARCHAR(100)	YES	
	country	VARCHAR(100)	YES	
	postal_code	VARCHAR(20)	YES	
	customer_class	VARCHAR(50)	YES	CustomerClass from source
	customer_segment	VARCHAR(50)	YES	CustomerSegment from source
	ic_type	VARCHAR(30)	YES	ID/passport type
	primary_phone	VARCHAR(30)	YES	Denormalized from CustomerPhone (IsPrimary=true)
	effective_date	DATE	NO	SCD2: row valid from date
	expiry_date	DATE	YES	SCD2: NULL means current row
	is_current	BOOLEAN	NO	SCD2: TRUE for active record

3.3 dim_account (SCD Type 2)

Tracks account-level attributes over time. SCD2 captures changes to interest rates, overdraft limits, and account type upgrades while preserving history for point-in-time accurate balance and fraud analysis.

Key	Column Name	Data Type	Nullable	Description
PK	account_key	INT	NO	Surrogate key
FK	account_number	VARCHAR(50)	NO	Business key from Account
FK	customer_key	INT	NO	FK to dim_customer
FK	branch_key	INT	YES	FK to dim_branch
	account_type	VARCHAR(50)	NO	Savings, Current, etc.
	currency	CHAR(3)	NO	ISO currency code
	open_date	DATE	YES	
	last_activity_date	DATE	YES	
	interest_rate	DECIMAL(8,4)	YES	
	minimum_balance	DECIMAL(18,2)	YES	
	overdraft_limit	DECIMAL(18,2)	YES	
	device_type	VARCHAR(50)	YES	Device used for digital banking
	ip_address	VARCHAR(45)	YES	Registered IP
	effective_date	DATE	NO	SCD2
	expiry_date	DATE	YES	SCD2
	is_current	BOOLEAN	NO	SCD2 current flag

3.4 dim_merchant

Merchant reference data. MCC (Merchant Category Code) is a key feature in fraud model scoring. Type 1 (overwrite) as merchant attribute changes are not critical to preserve historically.

Key	Column Name	Data Type	Nullable	Description
PK	merchant_key	INT	NO	Surrogate key
FK	merchant_id	VARCHAR(50)	NO	Business key
	merchant_name	VARCHAR(200)	NO	
	merchant_category	VARCHAR(100)	YES	
	mcc	VARCHAR(10)	YES	Merchant Category Code (ISO 18245)
	city	VARCHAR(100)	YES	
	state	VARCHAR(100)	YES	
	country	VARCHAR(100)	YES	
	postal_code	VARCHAR(20)	YES	
	business_type	VARCHAR(100)	YES	
	website_url	VARCHAR(500)	YES	
	latitude	DECIMAL(10,7)	YES	
	longitude	DECIMAL(10,7)	YES	
	manager_id	VARCHAR(50)	YES	

3.5 dim_card

Card attributes. CRITICAL: The raw card number is NEVER stored in the DWH. Only a SHA-256 hash is retained to satisfy PCI DSS requirements while still enabling card-level analytics and linkage to source system records.

Key	Column Name	Data Type	Nullable	Description
PK	card_key	INT	NO	Surrogate key
FK	card_number_hashed	VARCHAR(128)	NO	SHA-256 of card number (PCI DSS)
FK	account_key	INT	NO	FK to dim_account
	card_type	VARCHAR(50)	NO	Debit / Credit / Prepaid
	card_network	VARCHAR(50)	YES	Visa, Mastercard, Amex
	issue_date	DATE	YES	
	expiry_date_card	DATE	YES	Card expiry (not SCD field)
	card_holder_name	VARCHAR(200)	YES	
	daily_withdrawal_limit	DECIMAL(18,2)	YES	
	daily_purchase_limit	DECIMAL(18,2)	YES	
	card_status	VARCHAR(30)	NO	Active, Blocked, Expired

3.6 dim_branch

Branch location reference. Used by fact_transaction, fact_loan_snapshot, and dim_employee/dim_account for geographic hierarchies in reporting.

Key	Column Name	Data Type	Nullable	Description
PK	branch_key	INT	NO	Surrogate key
FK	branch_id	VARCHAR(50)	NO	Business key
	branch_name	VARCHAR(200)	NO	
	branch_location	VARCHAR(200)	YES	
	city	VARCHAR(100)	YES	
	state	VARCHAR(100)	YES	
	country	VARCHAR(100)	YES	
	open_date	DATE	YES	
	branch_type	VARCHAR(50)	YES	

3.7 dim_employee (SCD Type 2)

Tracks analyst and reviewer assignment history. SCD2 is important here because fraud alert review assignments and salary/position changes must be captured for audit and compliance reporting.

Key	Column Name	Data Type	Nullable	Description
PK	employee_key	INT	NO	Surrogate key
FK	employee_id	VARCHAR(50)	NO	Business key
	first_name	VARCHAR(100)	NO	
	last_name	VARCHAR(100)	NO	
	position	VARCHAR(100)	YES	
FK	department_key	INT	YES	FK to dim_department
FK	branch_key	INT	YES	FK to dim_branch
	hire_date	DATE	YES	
	salary	DECIMAL(18,2)	YES	
	effective_date	DATE	NO	SCD2
	expiry_date	DATE	YES	SCD2
	is_current	BOOLEAN	NO	SCD2 current flag

3.8 dim_department

Organizational hierarchy for branch and employee reporting. Type 1.

Key	Column Name	Data Type	Nullable	Description
PK	department_key	INT	NO	Surrogate key
FK	department_id	VARCHAR(50)	NO	Business key
	department_name	VARCHAR(100)	NO	
FK	manager_key	INT	YES	FK to dim_employee (manager)
FK	branch_key	INT	YES	FK to dim_branch
	country_code	CHAR(3)	YES	

3.9 dim_atm

ATM location and hardware reference for ATM-channel transaction analytics and geographic fraud pattern detection.

Key	Column Name	Data Type	Nullable	Description
PK	atm_key	INT	NO	Surrogate key
FK	atm_id	VARCHAR(50)	NO	Business key
	location_name	VARCHAR(200)	YES	

Key	Column Name	Data Type	Nullable	Description
	city	VARCHAR(100)	YES	
	country	VARCHAR(100)	YES	
	atm_type	VARCHAR(50)	YES	
	manufacturer	VARCHAR(100)	YES	
	model	VARCHAR(100)	YES	
FK	branch_key	INT	YES	FK to dim_branch
	latitude	DECIMAL(10,7)	YES	
	longitude	DECIMAL(10,7)	YES	
	atm_status	VARCHAR(30)	YES	Active, Offline, Maintenance

3.10 dim_transaction_type

Lookup table for transaction type classification. Denormalized category and description fields avoid repeated lookups in queries.

Key	Column Name	Data Type	Nullable	Description
PK	transaction_type_key	INT	NO	Surrogate key
FK	transaction_type_id	VARCHAR(50)	NO	Business key
	transaction_type_name	VARCHAR(100)	NO	
	category	VARCHAR(100)	YES	e.g. Payment, Transfer, Withdrawal
	description	VARCHAR(500)	YES	

3.11 dim_fraud_model

This dimension is introduced by the DWH — it does not exist in the source OLTP. It tracks every version of the ML fraud detection model deployed to production. By linking fact_fraud_detection to this dimension, analysts can compare performance across model versions, audit threshold changes, and measure the impact of retraining cycles.

Key	Column Name	Data Type	Nullable	Description
PK	model_key	INT	NO	Surrogate key
	model_name	VARCHAR(100)	NO	ML model name
	model_version	VARCHAR(50)	NO	Semantic version e.g. v2.3.1
	algorithm	VARCHAR(100)	YES	XGBoost, LGBM, Neural Net, etc.
	training_date	DATE	YES	Date model was trained
	deploy_date	DATE	YES	Date model went to production
	retire_date	DATE	YES	NULL if still active
	threshold_score	DECIMAL(6,4)	YES	Decision boundary used at deploy time
	is_active	BOOLEAN	NO	Current production model flag
	description	VARCHAR(500)	YES	Model notes and feature set summary

4. Fact Tables

4.1 fact_transaction (Core Transaction Grain)

Grain: one row per individual financial transaction. This is the central fact in the schema. It captures all transactional events including card payments, ATM withdrawals, transfers, and branch transactions. The is_fraud_confirmed column is the ground truth label — it is written back by the analyst review process and is the primary training signal for ML model retraining.

Key	Column Name	Data Type	Nullable	Description
PK	transaction_fact_key	BIGINT	NO	Surrogate key
	transaction_id	VARCHAR(50)	NO	Business key from source
FK	date_key	INT	NO	FK to dim_date (posting date)
FK	posting_date_key	INT	YES	FK to dim_date (value date)
FK	customer_key	INT	NO	FK to dim_customer
FK	account_key	INT	NO	FK to dim_account
FK	card_key	INT	YES	FK to dim_card (null if not card txn)
FK	merchant_key	INT	YES	FK to dim_merchant
FK	atm_key	INT	YES	FK to dim_atm (ATM txns)
FK	branch_key	INT	YES	FK to dim_branch
FK	transaction_type_key	INT	NO	FK to dim_transaction_type
	amount	DECIMAL(18,2)	NO	Transaction amount
	currency	CHAR(3)	NO	ISO currency
	amount_usd	DECIMAL(18,2)	YES	Normalized to USD using rate at txn time
	transaction_method	VARCHAR(50)	YES	Online, POS, ATM, Branch, etc.
	channel	VARCHAR(50)	YES	Mobile, Web, Branch, ATM
	transaction_location	VARCHAR(200)	YES	
	city	VARCHAR(100)	YES	
	country	VARCHAR(100)	YES	
	latitude	DECIMAL(10,7)	YES	
	longitude	DECIMAL(10,7)	YES	
	is_fraud_confirmed	BOOLEAN	YES	Ground truth label from human review
FK	fraud_label_date_key	INT	YES	FK to dim_date when labeled
	etl_load_timestamp	TIMESTAMP	NO	DWH load timestamp

4.2 fact_fraud_detection (ML Inference Output)

Grain: one row per ML model inference run against a transaction. If a transaction is scored by multiple model versions (e.g. during a champion/challenger test), multiple rows will exist for the same transaction_fact_key with different model_keys. This enables A/B model comparison directly in the warehouse.

Key	Column Name	Data Type	Nullable	Description
PK	fraud_detection_key	BIGINT	NO	Surrogate key
FK	transaction_fact_key	BIGINT	NO	FK to fact_transaction
	transaction_id	VARCHAR(50)	NO	Business key (for direct joins)
FK	detection_date_key	INT	NO	FK to dim_date
FK	customer_key	INT	NO	FK to dim_customer
FK	account_key	INT	NO	FK to dim_account
FK	model_key	INT	NO	FK to dim_fraud_model
	fraud_score	DECIMAL(6,4)	NO	Raw model probability score 0.0–1.0
	is_fraud_predicted	BOOLEAN	NO	TRUE if score >= threshold at detection time
	fraud_type	VARCHAR(100)	YES	Model-predicted fraud category
	risk_indicators	VARCHAR(500)	YES	Top features that drove the score
	confidence_score	DECIMAL(6,4)	YES	Model confidence / calibrated probability
	risk_indication	VARCHAR(50)	YES	Low / Medium / High / Critical
	action_taken	VARCHAR(100)	YES	Block, Flag, Pass, Review
	is_required_review	BOOLEAN	YES	Flagged for analyst review
	detection_timestamp	TIMESTAMP	NO	Exact time of model inference
	etl_load_timestamp	TIMESTAMP	NO	DWH load timestamp

4.3 fact_fraud_alert (Human Review & Ground Truth)

Grain: one row per fraud alert raised. This fact table is the cornerstone of the feedback loop. The is_confirmed_fraud column is the definitive human judgment — TRUE (confirmed fraud), FALSE (false positive), or NULL (pending review). This column is the label used to retrain and evaluate ML models. The resolution_minutes column enables SLA and analyst performance reporting.

Key	Column Name	Data Type	Nullable	Description
PK	fraud_alert_key	BIGINT	NO	Surrogate key
	alert_id	VARCHAR(50)	NO	Business key from source FraudAlert
FK	fraud_detection_key	BIGINT	NO	FK to fact_fraud_detection
FK	transaction_fact_key	BIGINT	NO	FK to fact_transaction
FK	customer_key	INT	NO	FK to dim_customer
FK	account_key	INT	YES	FK to dim_account
FK	alert_date_key	INT	NO	FK to dim_date (alert raised)
FK	review_date_key	INT	YES	FK to dim_date (reviewed)
FK	expiry_date_key	INT	YES	FK to dim_date (alert expiry)
FK	assigned_to_employee_key	INT	YES	FK to dim_employee (analyst)
FK	reviewed_by_employee_key	INT	YES	FK to dim_employee (reviewer)
	alert_type	VARCHAR(100)	YES	
	alert_severity	VARCHAR(30)	NO	Low / Medium / High / Critical
	fraud_score_at_alert	DECIMAL(6,4)	YES	Score snapshot when alert was raised
	alert_status	VARCHAR(50)	NO	Open, In-Review, Confirmed, Dismissed
	is_confirmed_fraud	BOOLEAN	YES	Ground truth: TRUE=Fraud, FALSE=False Positive, NULL=Pending
	resolution_minutes	INT	YES	Minutes from alert creation to resolution
	etl_load_timestamp	TIMESTAMP	NO	DWH load timestamp

4.4 fact_loan_snapshot (Monthly Periodic Snapshot)

Grain: one row per loan per month-end snapshot. Periodic snapshot fact tables are ideal for loan portfolios because they capture the state at regular intervals without requiring change-data-capture. Outstanding balance, interest rate, and payment schedules are captured at each snapshot date.

Key	Column Name	Data Type	Nullable	Description
PK	loan_fact_key	BIGINT	NO	Surrogate key
	loan_id	VARCHAR(50)	NO	Business key
FK	customer_key	INT	NO	FK to dim_customer
FK	branch_key	INT	YES	FK to dim_branch
FK	snapshot_date_key	INT	NO	FK to dim_date (monthly snapshot)
FK	start_date_key	INT	YES	FK to dim_date (loan start)
FK	end_date_key	INT	YES	FK to dim_date (loan end)
FK	next_payment_date_key	INT	YES	FK to dim_date
	loan_type	VARCHAR(100)	YES	
	loan_amount	DECIMAL(18,2)	NO	Original disbursed amount
	outstanding_amount	DECIMAL(18,2)	YES	Remaining balance at snapshot date
	interest_rate	DECIMAL(8,4)	YES	Annual interest rate
	loan_term_months	SMALLINT	YES	
	monthly_payment	DECIMAL(18,2)	YES	
	account_number	VARCHAR(50)	YES	Associated account

4.5 fact_account_balance_snapshot (Daily Periodic Snapshot)

Grain: one row per account per day. Captures end-of-day balance for trend analysis, anomaly detection (sudden large drops/spikes are fraud signals), and regulatory reporting. A normalized balance_usd column enables cross-currency portfolio aggregation.

Key	Column Name	Data Type	Nullable	Description
PK	balance_fact_key	BIGINT	NO	Surrogate key
FK	account_key	INT	NO	FK to dim_account
FK	customer_key	INT	NO	FK to dim_customer
FK	snapshot_date_key	INT	NO	FK to dim_date (daily snapshot)
	balance	DECIMAL(18,2)	NO	End-of-day account balance
	available_balance	DECIMAL(18,2)	YES	
	currency	CHAR(3)	NO	
	balance_usd	DECIMAL(18,2)	YES	Normalized balance in USD

5. Feedback Loop Integration

The schema is specifically designed to close the ML feedback loop. The following data flow describes how labels move from human review back to model retraining:

Step	Source	Action
1	OLTP → DWH	fact_transaction loaded via CDC/batch ETL. is_fraud_confirmed is initially NULL.
2	ML Pipeline → DWH	Model scores each transaction. fact_fraud_detection row inserted with fraud_score, model_key, and action_taken.
3	Alert Engine → DWH	If score >= threshold, fact_fraud_alert row created with alert_status = 'Open'.
4	Analyst → DWH	Analyst reviews alert and sets alert_status + is_confirmed_fraud. ETL writes confirmed label back to fact_transaction.is_fraud_confirmed.
5	DWH → ML Training	Retraining pipeline queries: SELECT transaction_id, features, is_fraud_confirmed FROM fact_transaction JOIN fact_fraud_detection WHERE is_fraud_confirmed IS NOT NULL. This produces a fully labeled training dataset.
6	New Model → dim_fraud_model	Retrained model registered in dim_fraud_model with new version, threshold, and deploy_date. Old model's retire_date is set. is_active flipped to new model.
7	Monitor	Compare fraud_score distributions and precision/recall across model_key values in fact_fraud_detection to detect drift and evaluate champion vs challenger.

6. Key Analytics Enabled by This Schema

Analytics Use Case	Tables Involved
Model Precision & Recall by Version	`fact_fraud_detection + fact_fraud_alert + dim_fraud_model`
False Positive Rate by Merchant Category	`fact_fraud_alert + dim_merchant + fact_transaction`
Customer Fraud Risk Score Trend	`fact_fraud_detection + dim_customer + dim_date`
Alert Resolution SLA Compliance	`fact_fraud_alert + dim_employee + dim_date`
Transaction Volume by Channel & Country	`fact_transaction + dim_date + dim_account`
Monthly Loan Portfolio Outstanding Balance	`fact_loan_snapshot + dim_customer + dim_date`
Account Balance Anomaly Detection	`fact_account_balance_snapshot + dim_account + dim_date`
Fraud Hotspot Geography (ATM/Merchant)	`fact_transaction + dim_atm + dim_merchant`
Model Drift: Score Distribution Over Time	`fact_fraud_detection + dim_fraud_model + dim_date`
Ground Truth Label Rate by Analyst	`fact_fraud_alert + dim_employee + dim_date`

7. Conventions & Best Practices

Naming: all tables use snake_case with dim_ and fact_ prefixes. All surrogate keys follow the pattern {table_name}_key. Business keys from source systems are preserved as {entity_name}_id columns alongside surrogate keys.

SCD2 Pattern: effective_date (NOT NULL), expiry_date (NULL = current), is_current (BOOLEAN). Current rows are always retrieved with WHERE is_current = TRUE. Point-in-time joins use BETWEEN effective_date AND COALESCE(expiry_date, '9999-12-31').

Security: card_number_hashed uses SHA-256. No raw card PANs, CVVs, or PINs are stored anywhere in the DWH. All PII columns should be access-controlled at the column level via row/column-level security in the target platform (Snowflake, BigQuery, Redshift, Databricks, etc.).

ETL Audit Columns: every fact table includes etl_load_timestamp to support incremental loads, data lineage, and debugging. Dimension tables include effective_date for SCD2 tracking.